

TITLE/INNOVATION NAME

ENERHIYA V.1: A HYBRID THERMO-KINETIC-PHOTOVOLTAIC WEARABLE SYSTEM

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

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INTRODUCTION

The rapid proliferation of Internet of Things (IoT) devices and wearable technology has created a significant demand for continuous, reliable power sources. From health monitors to smartwatches, modern electronics are heavily dependent on batteries that require frequent recharging, limiting their autonomy and user convenience. While advances in low-power electronics have improved efficiency, the "battery bottleneck" remains a critical challenge; devices often die when needed most, or users simply forget to charge them (Li, F., et al. (2022). Wearable energy systems: What are the limits and limitations? *National Science Review*, 9(11)).

Current solutions for extending battery life often rely on single-source energy harvesting, such as solar panels or kinetic chargers. However, these methods suffer from environmental intermittency—solar chargers fail at night or indoors, while kinetic chargers generate negligible power during periods of rest. There is a growing need for a power solution that is not dependent on a single environmental condition but can adapt to the user's daily life (Kim, J., et al. (2024). Advances in energy harvesting technologies for wearable devices. *Micromachines*, 15(7), 884.).

To address this, this study introduces **ENERHIYA**, a hybrid energy-harvesting wearable designed as a bracelet or watch strap. Unlike conventional power banks or single-source harvesters, ENERHIYA integrates three distinct technologies: thermoelectric generators (body heat), piezoelectric/electromagnetic transducers (motion), and photovoltaic cells (light). By combining these sources, the device aims to mitigate the inconsistency of individual energy harvesters, providing a stable, self-sustaining power source for portable electronics regardless of the time of day or user activity (Al-Mousawi, A. (2024). Designing an efficient hybrid energy harvesting system compatible with head/wrist-worn medical wearable devices. *Preprints*.).

The Need for Autonomous Wearable Power The evolution of wearable technology has shifted focus from simple data logging to continuous, autonomous sensing. However, the sustainability of these devices is often hindered by battery capacity. Research by Pritchard (2020) highlights the potential of harvesting energy from routine human activities, such as walking, to charge portable electronic devices, suggesting that the human body itself is an underutilized power station.

Kinetic and Ambient Energy Harvesting Kinetic energy harvesting has been identified as a primary method for powering wearables. Gottardi et al. (2016) explored kinetic harvesting specifically for the Internet of Things, demonstrating that movement can be converted into usable electricity to drive autonomous sensors. Furthermore, broader studies on ambient energy—including radio frequency and thermal gradients—indicate that the environment surrounding the user is rich in wasted energy that can be harnessed for sustainable operation..

The Shift to Hybrid Multi-Source Systems While individual harvesting methods show promise, they are limited by their environment. For instance, solar energy is useless without light, and kinetic energy drops to zero when the user is sedentary. To overcome this, recent scholarship advocates for hybrid systems. Wang et al. (2024) reviewed advances in harvesting technologies, noting that combining multiple energy transduction mechanisms is superior to single-source approaches..

This multi-source integration is the theoretical foundation of the ENERHIYA project. As noted by Jena et al. (2024), utilizing a hybrid approach—such as combining photovoltaics with piezoelectric and thermoelectric generators—allows for the compensation of one source's downtime with another's active time. This synthesis is crucial for ensuring that micro-watts of generated power accumulate into a viable supply, ensuring power delivery even in low-light or low-movement scenarios.

Specifically, the study seeks to address the following questions:

1. How can a wearable device be designed to physically integrate thermoelectric, piezoelectric, and photovoltaic harvesters into a single compact form factor (bracelet or watch strap)?
2. How can the system efficiently manage and combine diverse voltage inputs—fluctuating AC from kinetic motion versus DC from thermal and solar sources—into a single usable output?
3. How can the circuitry be optimized to minimize power conversion losses and ensure a stable voltage output (e.g., 5V) suitable for charging portable electronics?
4. How can the device be ergonomically designed to maintain necessary skin contact (for body heat harvesting) and light exposure (for solar harvesting) without compromising user comfort?
5. How effective and reliable is the hybrid prototype in maintaining battery charge levels under varying environmental conditions compared to single-source harvesters?

Therefore, this study aims to design and develop a **Hybrid Wearable Energy Harvesting Device** capable of harnessing **ambient energy** through **thermoelectric, piezoelectric, and photovoltaic technologies**, converting diverse inputs into usable electricity, and regulating power for portable electronics. The system seeks to assist mobile users in providing continuous, sustainable, and autonomous power, especially in remote, outdoor, or daily scenarios where conventional wall charging is inaccessible.

The main objective of this study is to design, develop, and test a **Smart Wearable Power Bank** that can simultaneously harvest body heat, kinetic motion, and ambient light. The device aims to address the limitations of single-source energy harvesters (intermittency) by integrating multi-source transduction mechanisms in one sleek, wrist-worn system, ensuring reliable power delivery even in low-light, indoor, or sedentary environments.

To support this main objective, the following specific objectives are set to be addressed:

- **Design and construct** a sleek, wearable prototype (bracelet or watch strap) that houses thermoelectric generators (TEGs), piezoelectric sensors, and flexible photovoltaic cells to harvest ambient energy.
- **Develop** a Power Management Unit (PMU) circuit capable of rectifying, regulating, and boosting the distinct voltage inputs from body heat, motion, and light into a unified charging current.
- **Implement** a hybrid storage interface that mitigates the intermittency of individual sources (e.g., utilizing solar during the day and thermal/kinetic energy at night or indoors) to ensure continuous power availability.
- **Integrate** the harvesting components into a chassis that maximizes the thermal gradient (temperature difference) and light absorption while ensuring the device remains lightweight and aesthetically pleasing for daily use.
- **Conduct** performance testing to measure the device's voltage output, charging efficiency, and stability under simulated scenarios (e.g., varying light intensities, physical activities, and ambient temperatures) to validate the superiority of the multi-source approach.

Scope of the Study This study focuses on the **design, development, and testing of "ENERHIYA," a hybrid wearable energy-harvesting device** intended to extend the battery life of portable electronics and IoT devices. The device is designed to harvest ambient energy from three distinct sources simultaneously: **thermoelectric generators (TEGs)** for body heat, **piezoelectric or electromagnetic transducers** for kinetic motion, and **photovoltaic cells** for ambient light.

The scope includes the integration of these transduction mechanisms into a **sleek, wrist-worn form factor (bracelet or watch strap)**. It involves the engineering of a **Power Management Unit (PMU)** capable of rectifying diverse voltage inputs (AC from motion, DC from thermal/solar), regulating the output to a stable level suitable for charging, and storing energy in a buffer battery. The study aims to demonstrate that a multi-source approach can mitigate the intermittency issues found in single-source harvesters (e.g., solar failing at night).

Delimitations of the Study The study is **delimited to low-power portable electronics** (e.g., smartwatches, fitness trackers, or smartphone trickle-charging); high-power applications such as laptops or heavy machinery are outside the scope of this research. The device is intended as an **auxiliary power source or range extender** rather than a complete replacement for grid-based charging, meaning it reduces charging frequency but may not eliminate the need for wall outlets entirely depending on usage intensity.

Testing and evaluation will be conducted through **simulated environmental scenarios** (e.g., using heat lamps to simulate body heat, shakers for motion, and controlled lighting for photovoltaics) rather than long-term field trials. The evaluation will focus on the device's **voltage output stability, conversion efficiency, and harvesting capacity** under varying conditions, rather than durability against extreme weather or water submersion. Additionally, the study is limited to a **prototype implementation** utilizing off-the-shelf harvesting components and does not cover the fabrication of custom semiconductor materials or mass production processes.

The development of **ENERHIYA** holds practical and societal importance, addressing the critical "battery bottleneck" that currently limits the autonomy of portable electronics and IoT devices. For mobile users, field professionals, and outdoor enthusiasts, the device serves as a reliable, autonomous power source that mitigates the anxiety of battery depletion in remote or resource-limited areas. By providing a continuous charge through ambient energy harvesting, it reduces reliance on conventional wall outlets and ensures that essential communication and monitoring devices remain operational when needed most.

For the general consumer and non-technical users, the device provides a sleek, user-friendly solution that integrates seamlessly into daily life as a bracelet or watch strap. This accessibility empowers individuals to adopt renewable energy practices without altering their daily routines. Furthermore, the device can serve as a practical educational tool in academic institutions, offering students and researchers hands-on insight into the physics of **hybrid energy harvesting**—demonstrating how thermal, kinetic, and solar energies can be effectively combined and managed.

From a technological perspective, this study contributes to the advancement of **green electronics** by proving the feasibility of a multi-source power management system in a wearable form factor. It addresses the limitations of single-source harvesters (which suffer from environmental intermittency) by demonstrating how to rectify and combine diverse voltage inputs—AC from motion and DC from heat/light—into a stable output. This innovation paves the way for future developments in self-powered sensors and the realization of truly autonomous Internet of Things (IoT) ecosystems.

The study also aligns with key **Sustainable Development Goals (SDGs)**. It supports **SDG 7: Affordable and Clean Energy** by promoting the use of ambient, renewable energy sources to power personal devices. It advances **SDG 9: Industry, Innovation, and Infrastructure** through the creation of innovative, hybrid energy technologies that enhance the sustainability of IoT systems. Furthermore, it contributes to **SDG 12: Responsible Consumption and Production** by potentially extending the lifespan of electronic devices and reducing the environmental waste associated with disposable batteries and frequent grid charging. Overall, this study is expected to enhance energy autonomy, promote sustainable technology habits, and foster innovation in the field of wearable electronic electronics.

Conceptual Framework

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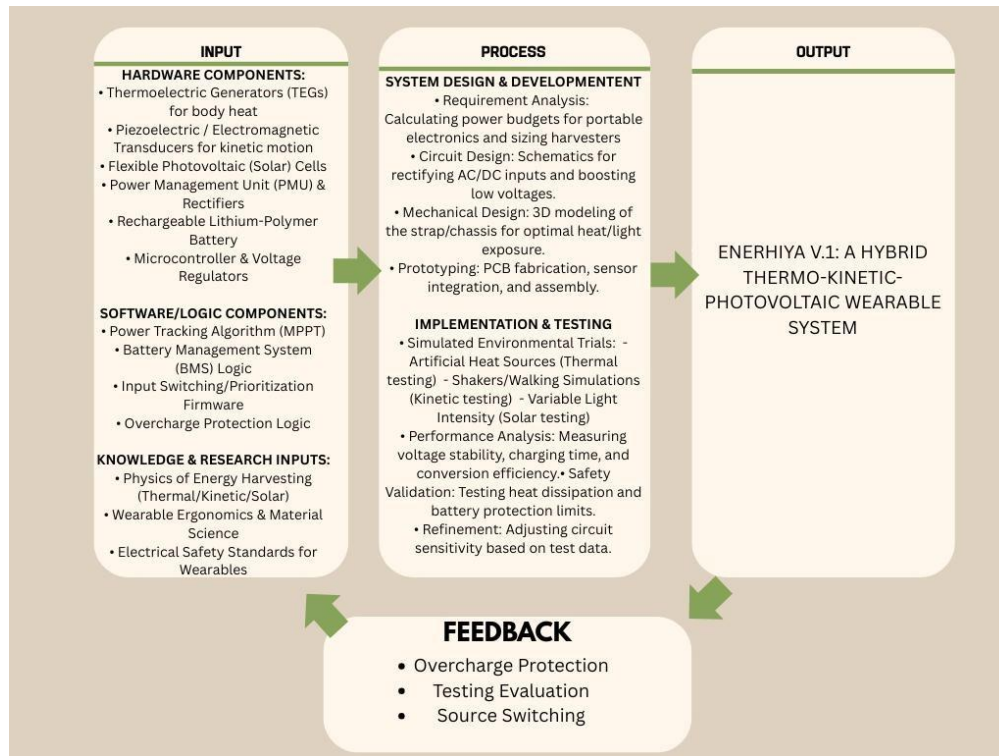


Figure 1. Conceptual Framework

The conceptual framework of **ENERHIYA** shows how the hybrid energy-harvesting system is developed from input to output. The **input** includes the main **hardware components** like the Thermoelectric Generators (TEGs) for body heat, Piezoelectric/Electromagnetic Transducers for kinetic motion, Flexible Photovoltaic (Solar) Cells, the Power Management Unit (PMU), the rechargeable Lithium-Polymer Battery, and the microcontroller. It also includes **software parts** such as the Power Tracking Algorithm (MPPT), the Battery Management System (BMS) logic, and the Input Switching/Prioritization Firmware necessary to combine diverse energy inputs. Research on the physics of energy harvesting, low-power circuit design principles, and wearable ergonomics also serve as an input reference.

The **process** involves designing, developing, and testing the device, starting from integrating the hardware, designing the power management circuit to rectify and boost low voltages, and assembling the prototype into a sleek wrist-worn chassis. This phase includes conducting **Simulated Environmental Trials** (like thermal, kinetic, and solar testing) to measure voltage stability, charging efficiency, and power output under varying conditions.

The **output** is the fully functional **ENERHIYA: Hybrid Wearable Energy Harvesting Device**, which is a self-sustaining power bank capable of autonomously harvesting energy from body heat, motion, and light to provide continuous and stable power for portable electronics. **Feedback** from performance testing and safety validation is used to refine the system's design, optimize the PMU algorithm, and adjust component sensitivity, ensuring long-term reliability and power efficiency.

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